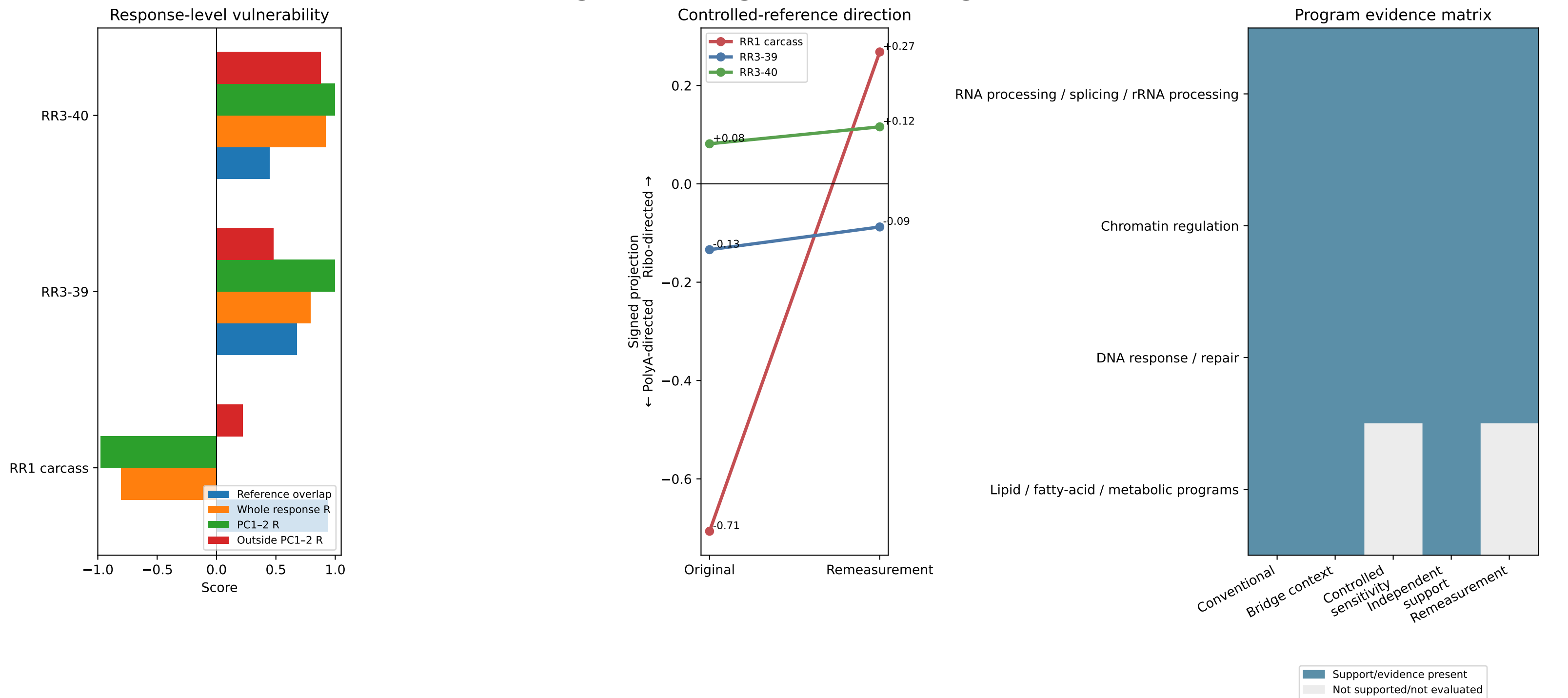


BridgeRNA Biological Confounding Profiler



Contextual organization beyond conventional DE: 596 RR1 genes are in the top 5% of positive expression-adjusted contextual residuals while edgeR FDR ≥ 0.05 . Existing enrichment finds 23 RNA-processing, 5 chromatin, and 2 DNA-repair significant terms. These are context-excess / non-significant-DE genes—not novel genes or genes missed by RNA-seq.

Interpretive principle: technical sensitivity identifies vulnerability, not artifact. RNA processing has biological and controlled technical support; chromatin/DNA programs recur independently but remain candidates rather than proven biology; lipid/metabolic programs recur broadly and lack controlled T-cell support. Global PC1-2 removal is inappropriate because RR3 carries highly reproducible response within the same reference.